

Customer Churn & Retention Analysis

Final Summary

1. Overview

The analysis examined customer churn patterns using contract type, tenure, monthly charges, payment method, selected services, and rule-based customer segments.

- **Total customers:** 7,043
- **Churned customers:** 1,869
- **Retained customers:** 5,174
- **Overall churn rate:** 26.54%

2. Key Findings

1. **Contract type:** Month-to-month customers have the highest churn rate at **42.71%**, compared with 11.27% for one-year and 2.83% for two-year contracts.
2. **Tenure:** Customers with **0–12 months of tenure** have the highest churn rate at **47.44%**, while customers with 49+ months have a rate of only 9.51%.
3. **Monthly charges:** Churned customers have higher average monthly charges (**\$74.44**) than retained customers (**\$61.27**).
4. **Payment and services:** Electronic-check customers have the highest payment-method churn rate (**45.29%**). Customers without Tech Support and Online Security also show high churn rates (**41.64%** and **41.77%**, respectively).
5. **Revenue exposure:** Churned customers account for **\$139,130.85 in monthly revenue at risk**.

3. Statistical Results

Test	Result	Decision	Conclusion
Contract vs Churn — Chi-Square	$\chi^2 = 1184.60$, $p < 0.05$	Reject H_0	Contract type and churn are significantly associated.
Monthly Charges vs Churn — T-Test	$t = 16.54$, $p < 0.05$	Reject H_0	Average monthly charges differ significantly between churned and retained customers.

4. Priority Customer Segment

New customers require the most retention attention.

Segment	Customers	Avg. Tenure	Avg. Monthly Charges	Churn Rate
New	2,186	4.73 months	\$56.10	47.44%
High-Risk	2,711	47.04 months	\$92.52	24.90%
High-Value	1,298	28.70 months	\$38.48	10.02%
Loyal	848	62.35 months	\$38.58	3.18%

The New segment has a churn rate of **47.44%**, substantially higher than all other segments, making early-stage customer retention the clearest priority.

5. Monthly Revenue at Risk

\$139,130.85 per month

This represents the combined MonthlyCharges of customers who have churned and therefore indicates the recurring monthly revenue currently lost through customer attrition.

6. Recommendations

1. **Strengthen early customer retention:** Focus onboarding, follow-ups, and support efforts on customers during their first 12 months.
2. **Encourage longer contracts:** Develop appropriate incentives or benefits to encourage month-to-month customers to move toward one-year or two-year contracts.
3. **Review high-charge customers:** Examine pricing and service combinations for customers with higher monthly charges and identify opportunities to improve perceived value.
4. **Investigate high-churn service/payment groups:** Examine the customer experience of electronic-check users and customers without Tech Support or Online Security, and develop targeted interventions where appropriate.
5. **Prioritize revenue-sensitive retention:** Direct retention resources toward customer groups where reducing churn can protect the greatest amount of recurring monthly revenue.